

Variability from Bounded Compression and Transversal Fixed Points

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Status. Candidate substantial partial result, likely valid. The bounded-compression theorem covers general BV coefficients for almost every initial point and the interface theorems give explicit fixed-point classes. The packet still does not replace the Doss transform for a general rough driver at every prescribed initial point.

Abstract

Hinz, Tölle, and Viitasaari ask for assumptions under which the integral process for a BV coefficient is itself variable. The main new observation is measure theoretic: if a family of trajectories has uniformly bounded compression, then for Lebesgue-almost every initial point its trajectory is $(s, 1)$ -variable with respect to every locally BV coefficient, simultaneously for all $s < 1$. Applied to the source’s bi-Lipschitz Doss flow, this removes all occupation assumptions on an arbitrary Hölder driver for almost every initial point. Applied to Ambrosio’s regular Lagrangian flow theorem, it gives a non-Doss result for bounded BV coefficients with bounded column divergences and Lipschitz drivers. We also retain two pointwise fixed-point theorems for a genuine hyperplane jump. A common normal row and a bi-Lipschitz driver clock yield Young–Picard solutions for rough drivers; a positive normal velocity yields a direct Schauder solution for absolutely continuous drivers. Both have the sharp conclusion $X \in V(\sigma, s, p)$ for $sp < 1$. An exact elliptic example shows that exceptional initial points and tangential collapse cannot simply be omitted.

1 Source question and answer

The source is Hinz, Tölle, and Viitasaari [1]. In Section 1.2, PDF page 6, after their Doss-transform existence theory, they write that the main open problem is to prove, under reasonable assumptions, that “the integral process itself will be variable,” with the goal of using standard fixed-point arguments.

This packet now gives two complementary advances. First, bounded compression forces variability for almost every starting point with no geometric regularity of the singular set. This applies to the full multidimensional Doss class of the source and, in a different time-regularity regime, to regular Lagrangian flows without Doss structure. Second, for one codimension-one jump interface it gives pointwise fixed-point constructions. The rough interface theorem keeps $Y \in C^\gamma$, $\gamma > 1/2$, while the absolutely continuous theorem allows a state-dependent normal velocity.

2 Definitions and setup

Let $H = \{x \in \mathbb{R}^n : h(x) = 0\}$, where

$$h(x) = \nu \cdot x - a, \quad |\nu| = 1.$$

Given two matrix fields $\sigma_\pm : \mathbb{R}^n \rightarrow \mathbb{R}^{n \times n}$, set

$$\sigma(x) = \begin{cases} \sigma_+(x), & h(x) \geq 0, \\ \sigma_-(x), & h(x) < 0. \end{cases} \quad (1)$$

fractional Brownian motions Y in $\mathbb{R}^n$ with $n \geq 2$ and Hurst index $H > \frac{1}{2}$. One can regard Corollary 3.26 as (a partial) extension of the probabilistic [89] Theorem 2.1]. Our fourth main result is a related uniqueness result, Theorem 3.28. It shows that Assumptions 3.12 and 3.15 guarantee uniqueness in the class of variability solutions.

It would be desirable to replace the Doss transform by standard fixed point arguments. The main open problem to be settled is to prove that — under reasonable assumptions — the integral process itself will be variable. Another goal for future research is to target equations that, in addition to a BV_{loc} -diffusion coefficient, involve a drift vector field of low regularity. First results on variability and compositions involving discontinuous paths can be found in [55].

1.3 Structure of the article

The structure of the article is as follows: In Section 2 we introduce the notion of (s, p) -variability, define compositions $\varphi(X)$ ($\sigma(X)$, respectively), and state our results on existence and properties of (1.1). We also provide a change of variable formula and a result

Figure 1: The source statement on PDF page 6. This is a direct crop of the source paper.

The representative chosen on H is irrelevant to all distributional derivatives and to the integrals below, because a solution meets H at most once.

For a scalar $\varphi \in BV_{\text{loc}}(\mathbb{R}^n)$, a continuous path $X : [0, T] \rightarrow \mathbb{R}^n$ is (s, p) -variable with respect to φ if some relatively compact open neighborhood U of $X([0, T])$ satisfies

$$t \mapsto \int_U |X_t - z|^{-n+1-s} \|D\varphi\|(dz) \quad \text{in } L^p(0, T).$$

For a matrix coefficient this is required componentwise, and we write $X \in \mathbb{V}(\sigma, s, p)$. The source notation $\mathbb{V}(\sigma, s)$ means $p = 1$.

We study

$$X_t = x_0 + \int_0^t \sigma(X_u) dY_u. \quad (2)$$

For Hölder drivers, integrals on intervals that do not contain the crossing are ordinary Young integrals. The global integral is their additive concatenation. It equals the source's generalized Lebesgue–Stieltjes integral: the integrand is Hölder on each side, has at most one jump in time, changing its value at the crossing does not alter either integral, and the restriction property joins the two pieces.

3 Idea of the proof

For bounded compression, the proof is a one-line interchange of integrals with a decisive consequence. If Φ_t pushes Lebesgue measure forward with density at most L , then for $\mu = |D\varphi| \upharpoonright_U$,

$$\int_B \int_0^T \int_U |\Phi_t(x) - z|^{-n+1-s} \mu(dz) dt dx \leq LT \int_U \int_K |y - z|^{-n+1-s} dy \mu(dz).$$

The last integral is finite because $n - 1 + s < n$. Hence the variability energy is finite for almost every initial point. Taking a countable sequence $s_j \uparrow 1$ gives one exceptional null set valid for every $s < 1$.

The source's Doss trajectories are conjugates of translations by a fixed bi-Lipschitz map, hence have bounded compression. Independently, Ambrosio's regular Lagrangian flows have bounded compression by definition. These two facts turn the abstract estimate into the generic-start results below.

The decisive quantity is the signed distance $h(X_t)$. A common-row identity for σ_+ and σ_- makes this coordinate independent of the unknown path:

$$\nu^\top \sigma_\pm(x) = \lambda^\top \implies h(X_t) = h(x_0) + \lambda \cdot (Y_t - Y_0).$$

If the right-hand side is a strictly increasing bi-Lipschitz clock, the side of the interface occupied at every time is known before solving the equation. One therefore solves an ordinary smooth Young equation with σ_- before the unique crossing and one with σ_+ afterward. Standard Picard contractions give the two pieces and uniqueness.

The same identity proves variability. Locally, the derivative measure of a piecewise smooth coefficient is bounded by volume measure plus surface measure on H . The volume contribution to the relevant Riesz potential is uniformly bounded. The surface contribution at a point x has size

$$\int_H |x - z|^{-n+1-s} d\mathcal{H}^{n-1}(z) \asymp \text{dist}(x, H)^{-s}.$$

Linear crossing turns this into $|t - \tau|^{-s}$ near the unique hitting time τ , and its p th power is integrable exactly when $sp < 1$.

For an absolutely continuous driver the normal velocity need not be a fixed row. It is enough that both one-sided vector fields have uniformly positive normal speed. Candidate paths with that same speed form a compact convex set. Although σ is discontinuous, uniform convergence of candidate paths gives pointwise convergence of the integrands away from the single crossing time. Dominated convergence makes the Picard operator continuous, so Schauder applies.

4 The bounded-compression principle

The next theorem has no interface or rectifiability hypothesis. It is the strongest result in the packet.

Theorem 1 (Bounded compression forces generic variability). *Let $B \subset \mathbb{R}^n$ be bounded and measurable and let $\Phi : [0, T] \times B \rightarrow \mathbb{R}^n$ be measurable, with each trajectory $t \mapsto \Phi_t(x)$ continuous. Suppose that*

$$(\Phi_t)_\#(\mathbf{1}_B \mathcal{L}^n) \leq L \mathcal{L}^n \quad (0 \leq t \leq T) \quad (3)$$

for a constant L , and that all the trajectories take values in one bounded set K . Let $\varphi \in BV_{\text{loc}}(\mathbb{R}^n)$. Then, outside one Lebesgue-null subset of B , the path $t \mapsto \Phi_t(x)$ is $(s, 1)$ -variable with respect to φ simultaneously for every $s \in (0, 1)$. The same conclusion holds componentwise for any finite matrix of locally BV functions.

Proof. Choose a bounded open set U containing the closure of K and put $\mu = |D\varphi| \llcorner_U$. For fixed $s \in (0, 1)$ define

$$E_s(x) = \int_0^T \int_U |\Phi_t(x) - z|^{-n+1-s} \mu(dz) dt.$$

Tonelli's theorem and (3) give

$$\int_B E_s(x) dx \leq L \int_0^T \int_U \int_K |y - z|^{-n+1-s} dy \mu(dz) dt. \quad (4)$$

If $K \cup U$ has diameter at most R , translation and polar coordinates give, uniformly in $z \in U$,

$$\int_K |y - z|^{-n+1-s} dy \leq |\mathbb{S}^{n-1}| \int_0^R r^{-s} dr = \frac{|\mathbb{S}^{n-1}| R^{1-s}}{1-s}.$$

Since $\mu(U) < \infty$, the right-hand side of (4) is finite. Thus $E_s(x) < \infty$ for almost every x , which is precisely $(s, 1)$ -variability.

To obtain a single null set, apply the preceding argument to a countable sequence $s_j \uparrow 1$. Given $s < s_j$, split distances at one; on the bounded set U the kernel of exponent $n - 1 + s$ is bounded by a constant times one plus the kernel of exponent $n - 1 + s_j$. Hence finiteness for s_j implies finiteness for s . A finite intersection handles all entries of a matrix coefficient. $\square$

Corollary 1 (Generic starts in the full Doss class). *Assume the multidimensional hypotheses of the source that produce a bi-Lipschitz Doss map $f : \mathbb{R}^n \rightarrow \mathbb{R}^n$ satisfying $\nabla f = \sigma \circ f$: namely its Assumptions 3.12 and 3.15 together with its angle condition (3.10). Let $Y \in C^\gamma([0, T]; \mathbb{R}^n)$ with $\gamma > 1/2$. Then for Lebesgue-almost every initial point $x \in \mathbb{R}^n$,*

$$X_t^x = f(Y_t - Y_0 + f^{-1}(x)) \quad (5)$$

belongs to $\mathcal{V}(\sigma, s)$ for every $s < 1$. Consequently it is the source's variability solution, and hence its unique variability solution under the source's uniqueness hypotheses. No upper-regularity, occupation-density, or Riesz-energy assumption on the driver is required.

Proof. Set $\Phi_t = f \circ \tau_{Y_t - Y_0} \circ f^{-1}$. Its inverse is $f \circ \tau_{Y_0 - Y_t} \circ f^{-1}$, so

$$\text{Lip}(\Phi_t^{-1}) \leq \text{Lip}(f) \text{Lip}(f^{-1})$$

uniformly in t . The standard measure estimate for a Lipschitz map gives (3) on every bounded set B , with $L \leq (\text{Lip}(f) \text{Lip}(f^{-1}))^n$. The range of $[0, T] \times B$ is bounded because $Y([0, T])$ and B are bounded and f is bi-Lipschitz. Theorem 1, first on $B(0, m)$ and then for all $m \in \mathbb{N}$, proves the simultaneous variability assertion.

This argument is not circular: formula (5) defines a continuous path before an integral is invoked. Bounded compression proves its variability. One may then follow the second half of the source's Lemma 5.7: the BV composition estimate for the bi-Lipschitz inverse transfers variability of X with respect to σ to variability of $Y - Y_0 + f^{-1}(x)$ with respect to each entry of $\nabla f = \sigma \circ f$. The source's Theorem 2.13 then applies the change-of-variables formula and identifies X with a solution of (2). The path is γ -Hölder. Choose $(1 - \gamma)/\gamma < s < 1$ to meet Definition 3.1 of the source. Its uniqueness theorem then applies whenever its stated Doss hypotheses are in force. $\square$

The same averaging lemma also combines with a completely different existence theory and thereby removes the Doss assumptions, at the price of a smoother driver and an almost-everywhere flow notion.

Corollary 2 (A non-Doss regular-Lagrangian-flow theorem). *Let $\sigma : \mathbb{R}^n \rightarrow \mathbb{R}^{n \times n}$ be bounded with locally BV entries and write $\sigma^{(k)}$ for its k th column. Assume*

$$\text{div } \sigma^{(k)} \in L^\infty(\mathbb{R}^n), \quad k = 1, \dots, n.$$

For $Y \in W^{1,\infty}([0, T]; \mathbb{R}^n)$, the vector field $b_t(x) = \sigma(x)\dot{Y}_t$ has a unique regular Lagrangian flow $X_t(x)$. For almost every x , its trajectory belongs to $\mathcal{V}(\sigma, s)$ simultaneously for every $s < 1$ and solves

$$X_t(x) = x + \int_0^t \sigma(X_u(x))\dot{Y}_u \, du.$$

In particular, regarding Y as C^γ for any $\gamma \in (1/2, 1)$ and choosing $s \in (1 - \gamma, 1)$, this trajectory is a variability solution in the sense of the source.

Proof. The field b belongs to $L_t^1 BV_{\text{loc}, x}$, is bounded, and

$$\text{div } b_t = \sum_{k=1}^n \dot{Y}_t^k \text{div } \sigma^{(k)} \in L^1(0, T; L^\infty(\mathbb{R}^n)).$$

Its boundedness supplies Ambrosio's growth condition. The regular Lagrangian flow theorem [2, 3] therefore gives existence, uniqueness in the regular-flow class, and a uniform compression bound. For almost every initial point its trajectory is absolutely continuous; discard the exceptional null set, which can only decrease each pushforward measure. For initial points in a bounded B , the estimate $|X_t(x) - x| \leq \|\sigma\|_\infty \|\dot{Y}\|_{L^1}$ places the trajectories in a common bounded set. Theorem 1 applies to every matrix entry. Finally the trajectories are Lipschitz, so in Definition 3.1 one may take path exponent $\alpha = 1$; the required inequality is then $s > 1 - \gamma$. $\square$

5 The rough-driver fixed-point theorem

We first recall the standard smooth Young equation in the precise strength needed here.

Lemma 1 (Young–Picard lemma). *Let $\gamma \in (1/2, 1)$, $Y \in C^\gamma([r, q]; \mathbb{R}^n)$, and $V \in C_b^2(\mathbb{R}^n; \mathbb{R}^{n \times n})$. For every $z \in \mathbb{R}^n$ there is a unique $Z \in C^\gamma([r, q]; \mathbb{R}^n)$ satisfying*

$$Z_t = z + \int_r^t V(Z_u) dY_u.$$

It is obtained by Picard contraction on sufficiently short subintervals and concatenation. If V is bounded, the concatenation reaches all of $[r, q]$.

Proof. For $f \in C^\gamma$ the Young–Loeve estimate gives, on an interval I ,

$$\left| \int_u^v f dY - f_u(Y_v - Y_u) \right| \leq C_\gamma [f]_\gamma [Y]_\gamma |v - u|^{2\gamma}.$$

On a C^γ ball of paths with a common initial value, boundedness of DV and D^2V gives

$$\begin{aligned} \|V(X) - V(Z)\|_\infty &\leq \|DV\|_\infty \|X - Z\|_\infty, \\ [V(X) - V(Z)]_\gamma &\leq C_V ([X - Z]_\gamma + ([X]_\gamma + [Z]_\gamma) \|X - Z\|_\infty). \end{aligned}$$

Since X and Z agree at the left endpoint, $\|X - Z\|_\infty \leq |I|^\gamma [X - Z]_\gamma$. The Young–Loeve estimate therefore makes the integral map a contraction when $|I|$ is small enough; the same estimate makes the chosen ball invariant. Iteration constructs the local solution. Uniqueness allows adjacent pieces to concatenate. The bounded vector field supplies a uniform local range bound, so finitely many pieces cover the compact time interval. $\square$

Theorem 2 (Rough clock-row theorem). *Let $\gamma \in (1/2, 1)$ and $Y \in C^\gamma([0, T]; \mathbb{R}^n)$. Suppose $\sigma_\pm \in C_b^2(\mathbb{R}^n; \mathbb{R}^{n \times n})$ and there is $\lambda \in \mathbb{R}^n$ such that*

$$\nu^\top \sigma_+(x) = \nu^\top \sigma_-(x) = \lambda^\top \quad (x \in \mathbb{R}^n). \quad (6)$$

Assume the scalar control

$$g(t) = h(x_0) + \lambda \cdot (Y_t - Y_0) \quad (7)$$

is increasing and bi-Lipschitz: for some $0 < m \leq M$,

$$m(t - u) \leq g(t) - g(u) \leq M(t - u), \quad 0 \leq u < t \leq T. \quad (8)$$

Then equation (2) has a unique $X \in C^\gamma([0, T]; \mathbb{R}^n)$ solution, constructed entirely by standard Young–Picard fixed points. Moreover,

$$X \in \mathbf{V}(\sigma, s, p) \quad \text{for every } s \in (0, 1), p \in [1, \infty) \text{ with } sp < 1. \quad (9)$$

In particular, choosing

$$\frac{1 - \gamma}{\gamma} < s < 1$$

makes X a variability solution for σ and Y in the exact sense of the source.

Proof. The strictly increasing function g has at most one zero. If it has a zero, call it τ ; otherwise the sign is constant. On every interval on which $g < 0$, solve the smooth Young equation with coefficient σ_- using Lemma 1; on every interval on which $g > 0$, use σ_+ . If there is a crossing, use the endpoint of the first solution as the initial value of the second.

Taking the scalar product with ν in either smooth equation and using (6) gives

$$h(X_t) = h(x_0) + \int_0^t \lambda \cdot dY_u = g(t).$$

Thus the constructed path lies in the half-space whose coefficient was used, so its concatenation solves the piecewise equation. The possible discrepancy at $t = \tau$ is a one-point change of the integrand and has zero integral.

Conversely, any solution of the piecewise equation has the same normal coordinate g . Its switching schedule is therefore identical. On each side it solves the corresponding smooth Young equation, where Lemma 1 gives uniqueness. This proves global uniqueness.

It remains to prove (9). Let $U \Subset \mathbb{R}^n$ be a bounded open neighborhood of $X([0, T])$. The BV product rule for (1) and the bounded first derivatives of the two branches give, for every scalar entry σ_{jk} ,

$$|D\sigma_{jk}| \llcorner U \leq C_U \mathcal{L}^n \llcorner U + J_U \mathcal{H}^{n-1} \llcorner_{H \cap U}. \quad (10)$$

The volume part has uniformly bounded potential because $s < 1$:

$$\sup_{x \in X([0, T])} \int_U |x - z|^{-n+1-s} dz < \infty.$$

For $x \notin H$, rotate and translate coordinates, enlarge $H \cap U$ to the whole hyperplane, and scale the tangential variable to obtain

$$\int_{H \cap U} |x - z|^{-n+1-s} d\mathcal{H}^{n-1}(z) \leq \int_{\mathbb{R}^{n-1}} (|w|^2 + |h(x)|^2)^{-(n-1+s)/2} dw \quad (11)$$

$$= C_{n,s} |h(x)|^{-s}. \quad (12)$$

For $n = 1$ this is the same identity with $\mathcal{H}^0$ on a point.

If g never vanishes, compactness gives $|g| \geq \delta > 0$. If it vanishes at τ , (8) gives

$$|h(X_t)| = |g(t)| \geq m|t - \tau|.$$

Combining this with (10) and (11) bounds each variability potential by $C(1 + |t - \tau|^{-s})$. Its p th power is integrable precisely when $sp < 1$; the infinite value at $t = \tau$ is immaterial. This proves (9).

Finally, X is γ -Hölder. Because $\gamma > 1/2$, the interval $((1 - \gamma)/\gamma, 1)$ is nonempty. For any s in it we have $X \in V(\sigma, s)$ and $\gamma s + \gamma > 1$, exactly the source definition of a variability solution. On each side the generalized integral agrees with the Young integral, and additivity across the single crossing proves that the constructed equation is the source equation. $\square$

Remark 1 (A concrete rough family). *After a linear change of coordinates, take $\nu = \lambda = e_1$ and $Y_t = (t, W_t)$, where W is any $(n - 1)$ -dimensional γ -Hölder path, $\gamma > 1/2$. Any two bounded C^2 branches whose first matrix row is $e_1^\top$ satisfy the theorem, regardless of the size of their jump in the remaining rows. Thus the theorem is not merely an ordinary smooth-time ODE: all tangential equations may be driven by a genuinely rough signal.*

6 A direct Schauder theorem for absolutely continuous drivers

The next theorem allows the normal row to depend on the point and to differ on the two sides. A common positive normal velocity replaces the algebraic identity (6).

Theorem 3 (Common-crossing Schauder theorem). *Let $Y \in W^{1,\infty}([0, T]; \mathbb{R}^n)$ and let the bounded fields $\sigma_\pm : \mathbb{R}^n \rightarrow \mathbb{R}^{n \times n}$ be locally Lipschitz. Assume there are constants $0 < c \leq M < \infty$ such that, for almost every t and every $x \in \mathbb{R}^n$,*

$$|\sigma_\pm(x) \dot{Y}_t| \leq M, \quad \nu \cdot \sigma_\pm(x) \dot{Y}_t \geq c. \quad (13)$$

Then the discontinuous integral equation

$$X_t = x_0 + \int_0^t \sigma(X_u) \dot{Y}_u du$$

has a unique absolutely continuous solution. It is obtained as a fixed point of the global discontinuous Picard map by the ordinary Schauder theorem, and it satisfies

$$X \in \mathcal{V}(\sigma, s, p) \quad (0 < s < 1, 1 \leq p < \infty, sp < 1).$$

Proof. Work in $C([0, T]; \mathbb{R}^n)$ and define K as the set of paths Z such that $Z_0 = x_0$ and, for $0 \leq u < t \leq T$,

$$|Z_t - Z_u| \leq M(t - u), \quad h(Z_t) - h(Z_u) \geq c(t - u). \quad (14)$$

The set is nonempty (it contains $x_0 + ct\nu$), convex, closed, uniformly bounded, and equicontinuous. Hence it is compact by Arzelà–Ascoli.

Define

$$(\mathcal{T}Z)_t = x_0 + \int_0^t \sigma(Z_u) \dot{Y}_u du.$$

The two inequalities in (13) show directly that $\mathcal{T}(K) \subset K$.

We claim that $\mathcal{T} : K \rightarrow K$ is continuous even though σ is not. If $Z^k \rightarrow Z$ uniformly in K , then $t \mapsto h(Z_t)$ is strictly increasing by (14); consequently Z meets H at at most one time. At every other time, Z_t lies strictly in one half-space, Z_t^k eventually lies in the same half-space, and local continuity of that branch gives

$$\sigma(Z_t^k) \dot{Y}_t \rightarrow \sigma(Z_t) \dot{Y}_t$$

for almost every t . The difference of the integrands is bounded by $2M$. Dominated convergence therefore yields

$$\|\mathcal{T}Z^k - \mathcal{T}Z\|_\infty \leq \int_0^T |\sigma(Z_t^k) \dot{Y}_t - \sigma(Z_t) \dot{Y}_t| dt \rightarrow 0.$$

Schauder’s theorem supplies a fixed point $X \in K$.

For uniqueness, every absolutely continuous solution automatically belongs to K . On each open time interval on which it stays in one half-space, it solves a time-dependent locally Lipschitz ODE. Two solutions agree by Gronwall until the first one reaches H ; since their paths then agree, their hitting times and hitting points agree. After the crossing they solve the same one-sided ODE from the same initial value. Thus they coincide globally.

Finally, (14) implies that $h(X_t)$ crosses zero at most once and, at a crossing time τ , $|h(X_t)| \geq c|t - \tau|$. The measure estimate (10) remains valid for locally Lipschitz branches on the bounded neighborhood U , and the argument (11) gives the stated (s, p) -variability. $\square$

7 Why transversality is doing real work

The conclusion is false if one keeps strong matrix nondegeneracy but drops output crossing. The companion packet proves the following exact example; we repeat its algebraic core to delimit Theorems 2 and 3.

Proposition 1 (Tangential-collapse obstruction). *In $\mathbb{R}^2$ let $H = \{x_2 = 0\}$ and*

$$A_+ = \begin{pmatrix} 2 & -1 \\ -1 & 1 \end{pmatrix}, \quad A_- = \begin{pmatrix} \frac{3}{2} & -\frac{1}{2} \\ -\frac{1}{2} & \frac{1}{2} \end{pmatrix}, \quad \sigma = A_+ \mathbf{1}_{\{x_2 \geq 0\}} + A_- \mathbf{1}_{\{x_2 < 0\}}.$$

The coefficient is bounded, symmetric, and uniformly positive definite; its inverse has distributionally curl-free rows, and it satisfies the determinant and angle assumptions in the source’s multidimensional Doss theory. For every $s \in (0, 1)$ and finite p , choose $0 < \alpha < \min\{1, (sp)^{-1}\}$ and set

$$X_t = (0, t^\alpha), \quad Y_t = (t, t).$$

Then $X \in \mathcal{V}(\sigma, s, p)$, but

$$\int_0^t \sigma(X_u) dY_u = (t, 0)$$

lies on H and is not (ρ, q) -variable for any $\rho > 0$ and any $q \geq 1$.

Proof. The eigenvalues of A_+ are $(3 \pm \sqrt{5})/2$ and those of A_- are $1 \pm 1/\sqrt{2}$. Both matrices are therefore uniformly positive definite. Their inverses are

$$A_+^{-1} = \begin{pmatrix} 1 & 1 \\ 1 & 2 \end{pmatrix}, \quad A_-^{-1} = \begin{pmatrix} 1 & 1 \\ 1 & 3 \end{pmatrix};$$

because they differ only in the $(2, 2)$ entry, the inverse rows have zero distributional curl across the horizontal line. Also $A_+(1, 1)^\top = A_-(1, 1)^\top = (1, 0)^\top$.

The jump measure is a nonzero multiple of $\mathcal{H}^1 \llcorner_H$. At distance t^α from H , its order- $(1-s)$ potential is $Ct^{-\alpha s}$, which is in L^p when $\alpha s p < 1$. The Picard image is $(t, 0)$ by the displayed matrix identity. On H the potential contains $\int_{-\varepsilon}^\varepsilon |r|^{-1-\rho} dr = \infty$ at every time, so the image is in no positive variability class. $\square$

Corollary 3 (The generic-start qualifier is sharp). *For the coefficient in Proposition 1, let $B \in C^\gamma([0, T])$ be nonconstant with $\gamma > 1/2$, put $Y = (B, B)$, and start at any $x \in H$. Then*

$$X_t = x + (B_t - B_0, 0)$$

is a solution that remains in H and is not (s, p) -variable for any $s > 0$ and $p \geq 1$. Moreover, there is no variability solution from such an initial point. Thus Theorem 1 cannot in general be improved from almost every start to every start, even for a uniformly elliptic coefficient satisfying the source's Doss structural conditions.

Proof. Both matrices satisfy $A_+(1, 1)^\top = A_-(1, 1)^\top = e_1$. On the displayed candidate the chosen representative is the constant matrix A_+ , so the Young integral is exactly $e_1(B_t - B_0)$ and the equation holds. Its range lies in H , where the Riesz potential of the nonzero jump measure is infinite at every time.

If a variability solution existed, all its component integrals would exist and their linearity, together with the same column-sum identity on both sides, would force it to equal the displayed candidate. That candidate is not variable, a contradiction. $\square$

Thus uniform ellipticity, smooth driving, and input variability do not make the Picard map preserve variability. The crossing hypotheses in the two main theorems prevent exactly the tangential collapse exhibited above.

8 Sharpness, limitations, and novelty

Theorem 1 is optimal in two immediate senses. The restriction $s < 1$ is exactly the local-integrability threshold for its Riesz kernel, and Corollary 3 exhibits an entire null hyperplane of initial points whose trajectories fail every positive variability condition. Bounded compression controls spatial averages; it cannot prohibit a singular exceptional orbit.

The threshold $sp < 1$ is sharp for a genuine jump and a linear crossing. If a component has nonzero jump trace at the hitting point, the hyperplane potential is comparable from below to $|t - \tau|^{-s}$, which is not in L^p when $sp \geq 1$. The $p = 1$ range nevertheless contains every $s < 1$, and hence the source compatibility interval $s > (1 - \gamma)/\gamma$ for every $\gamma > 1/2$.

The rough theorem deliberately assumes a single flat interface, smooth one-sided branches, a common normal row, and a bi-Lipschitz clock projection. The tangential driver can be fully Hölder-rough, but a driver with no monotone projection is not covered. Curved or multiple interfaces should be approachable with event functions and finitely many transversal hits, but their crossing schedule is no longer algebraically fixed and is not claimed here. The Schauder theorem allows state-dependent normal motion but requires an absolutely continuous driver.

The generic Doss corollary is broad in the coefficient and driver but still uses the Doss transform to construct the trajectory. The Ambrosio corollary does not use Doss and permits noninvertible coefficients, but it assumes a Lipschitz driver, bounded column divergences, and obtains a regular

Lagrangian flow only for almost every initial point. Therefore neither corollary is the requested general rough-driver fixed-point theorem at an arbitrary prescribed start. They instead identify bounded compression as a powerful sufficient mechanism for the authors’ central output-variability property.

A bounded search on 26 August 2026 covered the run’s lightweight indexes, the exact source phrase and arXiv id, the authors’ follow-ups, and combinations of “variability,” “fixed point,” “bounded compression,” “regular Lagrangian flow,” “hyperplane interface,” and “transversality.” No matching bounded-compression-to-variability theorem or fixed-point-plus-variability theorem was found. Ambrosio [2] supplies the regular Lagrangian flow theorem used in Corollary 2, but not the Riesz-energy averaging conclusion. Burden–Sastry–Koditschek–Revzen [4] prove well-defined piecewise-differentiable flows for event-selected discontinuous vector fields under transversality, but do not formulate the Riesz-potential variability conclusion or the rough clock-row theorem. López Pouso [5] uses Schauder ideas for discontinuous boundary-value problems; its null-contact-set continuity mechanism is philosophically close, but the equations and variability conclusion are different. The source itself proves that smooth curves with finitely many transversal hits are variable, but it does not use that fact to construct a fixed point. The present contribution is the combination: fixed-point construction, forced crossing geometry, and the sharp $sp < 1$ estimate.

Because the search is bounded rather than a systematic citation review, the proper label is *candidate substantial partial result, likely valid*. The bounded-compression result covers the source’s full Doss coefficient class only generically in the initial point, and the pointwise fixed-point results remain restricted to one interface. This is not a full answer to the source problem.

9 Verification checklist

The proof reduces to seven independently checkable mechanisms:

1. Tonelli plus bounded compression changes the initial-point average of the variability energy into an ordinary locally integrable Riesz kernel.
2. Bi-Lipschitz conjugates of translations have a uniform compression constant, so the estimate applies to the source’s Doss flow before its chain rule is used.
3. Ambrosio’s theorem applies to $b_t = \sum_k \dot{Y}_t^k \sigma^{(k)}$ under the stated bounded-divergence and growth hypotheses.
4. The common-row identity forces the normal coordinate of every solution to equal the prescribed clock g .
5. The fixed crossing schedule reduces the rough discontinuous equation to at most two standard smooth Young equations, each obtained by Picard contraction.
6. Surface measure on a hyperplane has Riesz potential exactly of order $\text{dist}(\cdot, H)^{-s}$; linear crossing gives the sharp condition $sp < 1$.
7. In the absolutely continuous case, a candidate path contacts the jump set at most once, which is enough for dominated convergence and continuity of the global Picard map.

No numerical evidence is used as proof.

References

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